

Chapter 3, Section 12

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1. Let Y be a scheme of finite type over an algebraically closed field k . Show that the function

$$\varphi(y) = \dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2)$$

is upper semicontinuous on the set of closed points of Y .

Proof. Apply (12.7.2) to the sheaf of relative differentials $\Omega_{Y/k}$. For each closed point $y \in Y$, we have a natural isomorphism (II, 8.7A)

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \cong \Omega_{Y/k} \otimes k(y).$$

□

2. Let $\{X_t\}$ be a family of hypersurfaces of the same degree in $\mathbf{P}_k^n$. Show that for each i , the function $h^i(X_t, \mathcal{O}_{X_t})$ is a constant function of t .

Proof. Hypersurfaces of same degree in projective space $\mathbf{P}^n$ have the same cohomology. For any hypersurface $X \subset \mathbf{P}^n$ of degree d , we have an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^n}(-d) \longrightarrow \mathcal{O}_{\mathbf{P}^n} \longrightarrow \mathcal{O}_X \longrightarrow 0,$$

and $h^i(X, \mathcal{O}_X)$ is independent of the choice of X .

□

4. Let Y be an integral scheme of finite type over an algebraically closed field k . Let $f : X \rightarrow Y$ be a flat projective morphism whose fibers are all integral schemes. Let $\mathcal{L}, \mathcal{M}$ be invertible sheaves on X , and assume for each $y \in Y$ that $\mathcal{L}_y \cong \mathcal{M}_y$ on the fiber X_y . Then show that there is an invertible sheaf $\mathcal{N}$ on Y such that $\mathcal{L} \cong \mathcal{M} \otimes f^*\mathcal{N}$.

Proof. If f is flat, then $\mathcal{L} \otimes \mathcal{M}^{-1}$ is flat over Y . For each $y \in Y$, $(\mathcal{L} \otimes \mathcal{M}^{-1})_y \cong \mathcal{O}_{X_y}$ on the fibre X_y , and each X_y is integral and projective over $k(y)$, so we have

$$h^0(X_y, (\mathcal{L} \otimes \mathcal{M}^{-1})_y) = h^0(X_y, \mathcal{O}_{X_y}) = 1.$$

Thus, $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank one on Y (12.9), say $f^*f_*(\mathcal{L} \otimes \mathcal{M}^{-1}) = \mathcal{N}$ for some $\mathcal{N} \in \text{Pic } Y$. Applying f^* to both sides, and using the fact that the natural morphism

$$f^*f_*(\mathcal{L} \otimes \mathcal{M}^{-1}) \rightarrow \mathcal{L} \otimes \mathcal{M}^{-1}$$

is actually an isomorphism because $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank one, we obtain our desired result.

□

5. Let Y be an integral scheme of finite type over an algebraically closed field k . Let $\mathcal{E}$ be a locally free sheaf on Y , and let $X = \mathbf{P}(\mathcal{E})$. Then show that $\text{Pic } X \cong (\text{Pic } Y) \times \mathbf{Z}$. This strengthens (II, Ex. 7.9).

Proof. Let n be the rank of $\mathcal{E}$, $\mathcal{O}(1)$ the canonical invertible sheaf of X , and $\pi : X \rightarrow Y$ the natural projection morphism. There is an obvious map

$$\begin{aligned} \varphi : (\text{Pic } Y) \times \mathbf{Z} &\rightarrow \text{Pic } X \\ (\mathcal{L}, d) &\mapsto \mathcal{O}(d) \otimes \pi^* \mathcal{L}. \end{aligned}$$

Since π is a flat projective morphism, we can apply the results of this section. In particular, if $\mathcal{M}$ is an invertible sheaf on X , then $\mathcal{M}_y$ is an invertible sheaf on $X_y \cong \mathbf{P}_y^n$ for all $y \in Y$, where $\mathbf{P}_y^n$ is the projective n -space over $k(y)$. In particular, we can write $\mathcal{M}_y \cong \mathcal{O}_{\mathbf{P}_y^n}(d_y)$ for some $d_y \in \mathbf{Z}$. We will show d_y is constant for all $y \in Y$. It will follow by (Ex. 12.4) that there exists an invertible sheaf $\mathcal{N}$ on Y such that $\mathcal{M} \cong \mathcal{O}(d) \otimes \pi^* \mathcal{N}$ for some $d \in \mathbf{Z}$, proving surjectivity of φ .

We proceed by showing the minimum value of d_y is obtained when $y = \xi$ is the generic point of Y . Suppose there exists $y' \in Y$ such that $d_{y'} < d_\xi$. After twisting by $\mathcal{O}(-d_{y'})$, we can assume $d_{y'} = 0$. By (12.8), the function

$$h^0(y, \mathcal{M}) = \dim_{k(y)} H^0(\mathbf{P}_y^n, \mathcal{O}_{\mathbf{P}_y^n}(d_y))$$

is upper semicontinuous on Y (12.8), and we have the strict inequality

$$\dim_{k(y')} H^0(\mathbf{P}_{y'}^n, \mathcal{O}_{\mathbf{P}_{y'}}^n) < \dim_{k(\xi)} H^0(\mathbf{P}_\xi^n, \mathcal{O}_{\mathbf{P}_\xi^n}(d_\xi))$$

by the explicit calculations of (5.1). But this implies the set $\{y \in Y \mid h^0(y, \mathcal{M}) \geq h^0(\xi, \mathcal{M})\}$ is a closed set containing ξ and not containing y' , a contradiction. Hence, we have $d_\xi \leq d_y$ for all $y \in Y$.

Next, we prove the inequality in the opposite direction. By Serre duality (5.1) (7.7), we have

$$\begin{aligned} h^n(y, \mathcal{M}) &= \dim_{k(y)} H^n(\mathbf{P}_y^n, \mathcal{O}_{\mathbf{P}_y^n}(d_y)) \\ &= \dim_{k(y)} H^n(\mathbf{P}_y^n, \mathcal{O}_{\mathbf{P}_y^n}(-d_y - n - 1)) \\ &= h^0(y, \mathcal{O}(-n - 1) \otimes \mathcal{M}^{-1}), \end{aligned}$$

and $h^n(y, \mathcal{M})$ is also an upper semicontinuous on Y . Suppose there exists $y' \in Y$ such that $d_\xi < d_{y'}$. After twisting by $\mathcal{O}(-d_\xi + n + 1)$, we can assume $d_\xi = n + 1$. In particular $d_{y'}$ is a positive integer. Again, we have the strict inequality

$$\dim_{k(y')} H^0(\mathbf{P}_{y'}^n, \mathcal{O}_{\mathbf{P}_{y'}}^n(-d_{y'} - n - 1)) < \dim_{k(\xi)} H^0(\mathbf{P}_\xi^n, \mathcal{O}_{\mathbf{P}_\xi^n}),$$

arriving at the same contradiction as above. Hence, d_y is constant for all $y \in Y$, and we have shown φ is surjective.

It remains to show φ is injective. If we have an isomorphism

$$\mathcal{O}(d) \otimes \pi^* \mathcal{L} \cong \mathcal{O}(d') \otimes \pi^* \mathcal{L}',$$

then it is clear by checking on fibres that $d = d'$, so it suffices to check $\pi^* : \text{Pic } Y \rightarrow \text{Pic } X$ is injective. Indeed, we have $\pi_* \mathcal{O}_X = \mathcal{O}_Y$ (II, 7.11a), and the projection formula gives

$$\pi_* \pi^* \mathcal{N} \cong \pi_* \mathcal{O}_X \otimes \mathcal{N} \cong \mathcal{N}$$

for all $\mathcal{N} \in \text{Pic } Y$, so if $f^* \mathcal{N} \cong \mathcal{O}_X$, then

$$\mathcal{N} \cong \pi_* \mathcal{O}_X \cong \mathcal{O}_Y.$$

□

6. Let X be an integral projective scheme over an algebraically closed field k , and assume that $H^1(X, \mathcal{O}_X) = 0$. Let T be a connected scheme of finite type over k .

- (a) If $\mathcal{L}$ is an invertible sheaf on $X \times T$, show that the invertible sheaves $\mathcal{L}_t$ on $X = X \times \{t\}$ are isomorphic, for all closed points $t \in T$.
- (b) Show that $\text{Pic}(X \times T) = \text{Pic } X \times \text{Pic } T$.

Proof.

- (a) Let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X .
- (b)

□